Random No.					Personal No.		

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P530/2

BIOLOGY

PRACTICAL

Paper 2

2026

3¼ hours

INVESTIGATIVE BIOLOGY PRE-UNEB SET 1

Uganda Advanced Certificate of Education

BIOLOGY PRACTICAL

3 hours 15 minutes

For Scorers' Use Only

Item No.	Code	Weighted Score	Scorers' Initials
1			
2			
Total			

INSTRUCTION TO CANDIDATES:

- ✓ *This examination paper consists of* **two**
- ✓ *items. Respond to* **both** *items.*
- ✓ *Write all responses in the spaces provided.*
- ✓ *Use sharp* **pencils** *for drawings. Coloured pencils should* **not** *be used.*
- ✓ *Show all your working for any calculation.*
- ✓ **No** *extra paper is allowed. Any work written on additional sheets will* **not** *be scored.*

Item 1

In Kasese District, Western Uganda, ecological researchers monitored two terrestrial vertebrates co-existing in a dry bushland area: a local agamid lizard and a wild bush rat.

During an unusually severe seasonal drought accompanied by strong winds, all soft-stemmed weeds were destroyed, and only tough, fibrous weeds survived. The lizard relied mainly on soft plant tissue for food and moisture, whereas the bush rat could switch its diet to fibrous plant stems and roots. Within three weeks after the storm, local officers reported a sharp decline in lizard populations, while bush rats maintained a stable, high population size.

To investigate the physiological and structural causes of this survival difference, ecologists analysed some parameters of both species.

The researchers recorded complete data for **Specimen P,** a lizard, but their study was interrupted before they could complete measurements for Specimen Q, a bush rat.

Table 1

Biological Parameter	**Specimen P**	**Specimen Q**
Dental Formula	$i\frac{5}{5}; c\frac{0}{0}; pm\frac{0}{0}; m\frac{0}{0}$	
Total Lung Surface Area (mm^2) measured by spreading lungs on graph paper	125	
Major routes of Blood flow from the gut to the heart for further distribution	Absent	
Length of the hindgut (both caecum and colon) in mm	40	

You are provided with a freshly killed **Specimen Q**, and two stem internodes from soft and fibrous plants labelled **K** and **L** respectively.

Task:

(a) Obtain data from **Specimen Q** to complete **Table 1**.

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(b) Make a drawing showing the exposed abdominal structures of Specimen Q and label them.

(c) Examine stems K and L using a microscope to identify tissues responsible for their different behaviours under strong winds. Draw and label relevant tissues.

(d) Analyse the findings from the investigation above to account for

(i) The difference in population sizes between specimens P and Q in their habitat during the environmental stress.

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(ii) The difference in the resistance of plants from which stems K and L were obtained to strong winds

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Item 2

A food processing cooperative in the Elgon region extracts natural pigments for food colouring from a coloured plant. During transport and processing, farmers noticed severe pigment loss when crop containers were exposed to heat from solar driers and when tubers were soaked in water contaminated with acidic chemical effluents from nearby processing mills. Experts in food processing believe that temperatures above 45^0C and contaminated water damage cell membranes and cause leakage of most of their contents. Colour intensity can be used to compare leakage levels when a peeled cylinder or strip is immersed in water for at least 10 to 15minutes.

You are provided with:

- **Tuber X,** from which the coloured substances are extracted
- **Solution A,** which is contaminated water with the effluents
- **Solution B,** which is uncontaminated water

Task:

Design and carry out a scientific investigation to show how prevailing factors during transportation and processing affect pigment loss and advise the cooperative on handling conditions for better output. *(Write your response in a scientific report)*

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ADVANCE INSTRUCTIONS TO TEACHERS

Great care should be taken that the information given below does not reach the candidates.

In addition to standard biological laboratory fittings, each candidate will require:

1. **Specimen Q:** One freshly killed mature rat (*Rattus norvegicus* or *Rattus rattus*), presented on a dissecting board with pins.
2. Beetroot Tissue labelled **X**
3. **K-** Young stem of commelina
4. **L-** Young stem of sunflower or young stem part of Lantana camara
5. **Solution A:** 25 cm^3 of 2.0 M Hydrochloric Acid solution in a beaker, labelled **Solution A**.
6. **Solution B:** 25 cm^3 of Distilled water in a beaker, labeled **Solution B**.
7. **Distilled Water:** 100 cm^3 in a wash bottle, labelled **Distilled Water**.
8. **Phloroglucinol and conc. HCl**
9. **Apparatus:**
    - 8 Test tubes in a test tube rack.
    - 10 cm^3 Graduated measuring cylinders (2 per candidate).
    - Thermometer
    - Beakers (250 cm^3) for water baths (hot water supply)
    - Stopwatch / digital timer.
    - Forceps / mounted needle.
    - Dissecting kit (scalpel, scissors, forceps, dissecting pins, dissecting board).
    - Light microscope with microscope slides.

P530/2 BIOLOGY PRACTICAL

SCORING GUIDE — Pre-UNEB Set 1

Item 1 = 12. Item 2 = 17.

ITEM 1

Code DS — Dissection procedures

Expected response

1. *Open the jaws and cut along the lateral line between the lower and upper jaws on both sides to expose the teeth.*
2. *Place the specimen with the ventral side uppermost and pin it through the feet.*
3. *Lift the skin on the mid-ventral line and cut from the lower jaw to the pelvic region.*
4. *Loosen the skin and pin it back.*
5. *Cut the abdominal wall on the midline from the diaphragm to the pelvic girdle and pin the wall back.*
6. *Displace the liver forward, the ileum to the left side; the duodenum to the right so that the hepatic portal vein and hepatic vein are displayed; and*
7. *The caecum and colon are displaced downwards to the right to expose them.*
8. *Cut out the alimentary canal by ligaturing the coeliac artery, hepatic portal vein, anterior mesenteric artery, and cut them distal to ligatured points. Remove the alimentary canal, excluding the hindgut, to expose the posterior vena cava.*
9. *Open the thorax by cutting along lateral lines of the rib cage and transversely along the posterior end of the rib cage to remove the ventral part of the rib cage completely.*
10. *Cut out the pericardium membrane and displace the heart anteriorly to the right to display the left lung; displace the right lung posterolaterally to the right to display the 4 lobes; and spread both lungs flat on graph paper.*

Note: *rectum has been ignored as part of hind gut. However, if a candidate includes it, ignore it*

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	**4≤** display or displacement steps, including displacement that shows the hepatic portal vein and the caecum and colon.	3
Ae	Two or three display or displacement steps, at least one of which moves an organ aside.	2
Al	One display step, or only orientation, pinning, skinning, and opening the body wall.	1
0	No procedure, or the cavity of Q is not opened.	0

Code DM — Measurements

Expected response

Biological Parameter	Specimen P	Specimen Q
Dental Formula	$I\frac{5}{5}; c\frac{0}{0}; pm\frac{0}{0}; m\frac{0}{0}$	$I\frac{1}{1}; c\frac{0}{0}; pm\frac{0}{0}; m\frac{3}{3}$
Total Lung Surface Area (mm^2) measured by spreading lungs on graph paper	125	*1600-2200*
Major routes of Blood flow from the gut to the heart for further distribution	Absent	*Hepatic portal vein, hepatic vein and posterior vena cava present*
Length of the hindgut (both caecum and colon) in mm	40	*80-120*

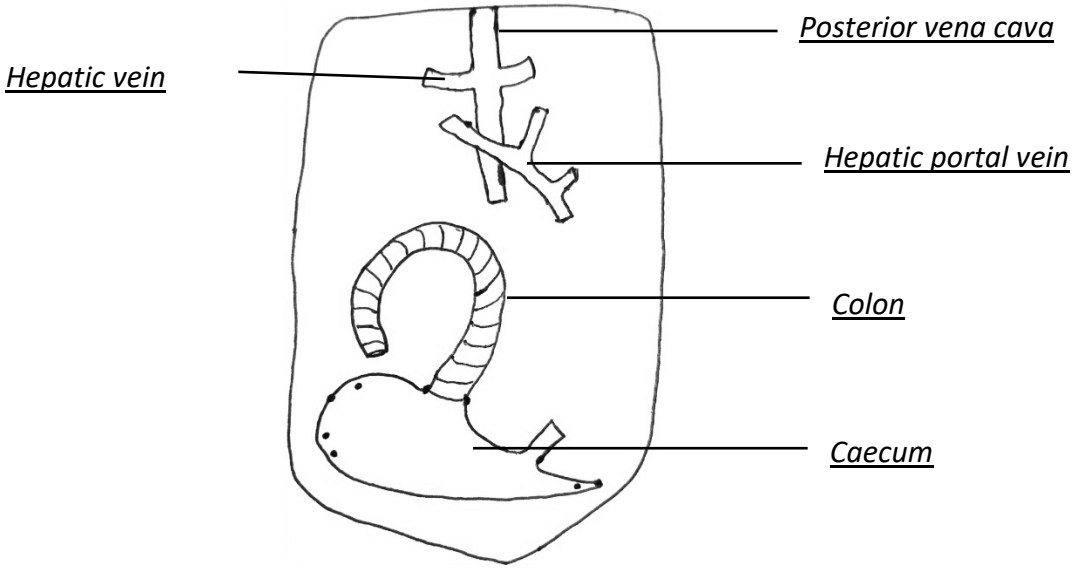
Bases of assessment

Level	What is seen on the script	Weighted score
Ap	Three or more of the four entries are correct and comparable with the value for P. Consider a candidate who mentions only the hepatic portal vein alone.	3
Ae	Exactly two entries are correct and comparable.	2
Al	Exactly one entry is correct and comparable.	1
0	No entry is correct, or values were copied from P.	0

Code DR — Drawing

Expected response

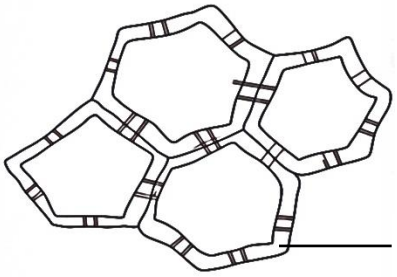
(b) A drawing showing the routes of blood from the gut to the heart (the hepatic vein, hepatic portal vein, posterior vena cava) and hindgut of specimen Q



X1-X3

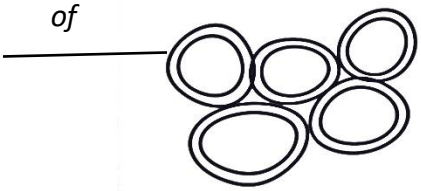
(c)

Drawing showing structures of K responsible for the behaviours of the plants in strong winds or Drawing showing sclerenchyma tissue



X150-X400

Drawing showing structures of K responsible for the behaviours of the plants in strong winds or drawing showing parenchyma tissue



X150-X400

Bases of assessment

Attributes per drawing are Title, Magnification, Accuracy, Labelling. Three drawings give 12 attributes.

Level	What is seen on the script	Weighted score
Ap	8 or more of the 12 attributes present.	3
Ae	5 to 7 attributes.	2
Al	1 to 4 attributes.	1
0	No drawing or coloured pencil used on any drawing.	0

Code ST — Suitability

(d) Expected response

- *Q has molars with broad and rough surfaces that provide a large and rough surface for grinding the fibrous stems; P has only one tooth type, for cutting and holding soft stems.*
- *Q's hindgut is longer than that of P, so microbial digestion of fibre takes longer than in P, thus extracting sufficient amounts of nutrients.*
- *Q has a longer hindgut, which provides larger space to accommodate larger amounts of symbiotic bacteria that secrete enzymes for efficient digestion of the fibrous structures, unlike the shorter hindgut in P, which provides smaller space/surface area for bacteria.*
- *The lung area of lungs of Q is above that of P, which presents a larger surface area for uptake of oxygen, thus maximising energy release from aerobic respiration.*
- *The hepatic portal vein, hepatic vein and posterior vena cava is present, so delivery of absorbed nutrients to the liver exceeds that of P, where that route is absent.*
- *Stem L has lignified sclerenchyma cells with thick walls that provide stronger support that resists bending and breaking to continue providing food to organisms Q, while stem K is filled with thin-walled, non-lignified parenchyma cells in the cortex that can easily collapse under strong winds, causing food shortage to the animal P.*

Bases of assessment

Any three of these four links.

Level	What is seen on the script	Weighted score
Ap	Three links, each with a datum or a drawn tissue and its significance.	3
Ae	Two such links.	2
Al	One such link.	1
0	Story restated with no datum and no drawn tissue.	0

ITEM 2

Code PE — Planning of the experiment

Expected response

Aim. *To find which wash liquid,* ***solution A or solution B****, and which* ***temperature, above 45 °C or below it,*** *gives the* ***least colour intensity*** *from a peeled cube of tuber X after 10 to 15 minutes of soaking,* ***to advise the cooperative on handling conditions to reduce pigment loss.***

Hypothesis. ***Solution A and a temperature above 45 °C*** *give* ***higher colour intensity*** *than* ***solution B and a lower temperature below 45^{0}C****, because acidic solution* ***denatures membrane proteins*** *and high temperature* ***disrupts the phospholipid bilayer,*** *both of which increase membrane leakage.*

Variables.

Independent: *Temperature; wash liquid (A or B) or pH of solutions A and B;*

Dependent: *Colour intensity.*

Controlled: *cylinder size (length and diameter); volume of liquid A and B; time of soaking;*

Level	What is seen on the script	Weighted score
2Ap	At least 2Ap among aim, hypothesis, and variables, and at least three accurate points.	4
1Ap	1Ap among aim, hypothesis, and variables, and at least three accurate points.	3

Ae	No Ap. At least one Ae (what is done without purpose, or two predictions without reasons).	2
Al	No Ap and no Ae. Only one independent variable, or one prediction with one reason.	1
0	Aim, hypothesis, and variables absent or not taken from the scenario	0

Code ME — Methods of the experiment

Expected response

Risks and mitigations

- *Solutions can burn the skin and destroy cell membranes of skin cells if spills over to skin during mixing; this is mitigated by wearing gloves and rinsing if it touches the skin.*
- *Hot Water with above 45 °C can scald the body if it spills while preparing a water bath or being poured into a test tube; it can be mitigated by use of holders.*

Procedure.

Effect of solutions A and B

1. *$10cm^3$ of solutions A and B were measured and added to the respective labelled test tubes A and B.*
2. *5cm cylinders from tuber X were cut using a cork borer of 0.5cm diameter, and each was placed in test tubes A and B.*
3. *A stopwatch was immediately started and set up for 10 minutes, after which colour intensity was observed and recorded.*

Effect of temperature

4. *Two test tubes were labelled 25^0C (any temperature below 45^0C) and 60^0C (any temperature above 45^0C)*
5. *$10cm^3$ of solution B or distilled water at 25^0C and 60^0C were measured and added to the respective test tubes. Tubes were incubated in baths at the same temperature to avoid temperature changes.*
6. *Procedures 2 and 3 were repeated using test tubes labelled 25^0C and 60^0C instead of A and B.*

Bases of assessment

- Relevant steps in the use of X, A, B, and the temperature contrast to obtain results.
- Accuracy is stated as **equal size**, **equal volume**, **thermometer/stated temperature**, and the **stated time frame**.
- Coherency is the order: **measure volume, cut tuber, set temperature, add tuber to solutions, wait, record.**

A source of error is not a risk.

If accurate points or coherent points are below 3, Ae is the maximum for that parameter.

Level	What is seen on the script	Weighted score
Ap	2Ap or more across relevance, accuracy, and coherency.	4
1 Ap	Exactly one Ap, with the rest Ae or Al.	3
Ae	No Ap. Risks paired with mitigations, or two accurate or two coherent points.	2
Al	One risk with a mitigation, or one accurate point, or one coherent link.	1

Code DA — Data presentation

Expected response

Test tubes	Observations
A	*Highly intense (colour) red solution/ dark red solution*
B	*Low-intensity (colour) red solution/ pale red solution*
25^0C	*Low-intensity (colour) red solution/ pale red solution*
60^0C	*Highly intense (colour) red solution/ dark red solution*

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	3 correct observations and above	3
Ae	Two correct observations	2
Al	One correct observation	1
0	No results or wrong results	0

Code AN — Explanation of results

Expected response

- *Temperatures above 45 °C damage the cell membrane of tissue X more than lower temperatures. At high temperatures, the phospholipids gain more kinetic energy, so the membrane becomes more fluid than at temperatures below 45°C, creating gaps between phospholipids. The same heat also denatures membrane proteins by breaking hydrogen bonds and hydrophobic interactions. The membrane therefore loses its normal structure, it becomes more permeable, and the coloured cell contents rapidly diffuse out. This makes the external solution turn intense red than at lower temperatures which provide less kinetic energy to phospholipids and do not denature membrane proteins.*
- *Solution A damages the membrane of cells of tissue X more than solution B because A was contaminated with acid effluent. The hydrogen ions in A disrupt hydrogen and ionic bonds in the membrane proteins which are denatured. Permeability increases, and more coloured cell contents leak into A than into the uncontaminated solution B, which leaves membrane proteins intact.*

The deepest colour is therefore A above 60 °C, which is the loss the farmers reported.

Level	What is seen on the script	Weighted score
Ap	2/3 of explanation points	3
Ae	1/3 of explanation points	2
Al	Below 1/3 of explanation points	1
0	No explanation.	0

Code CR — Conclusion and recommendation

Expected response

Conclusion.

Solution A and a temperature above 45 °C release more pigment than solution B and a temperature lower than 45 °C, because they destroy cell membranes, so the hypothesis is supported.

Recommendation.

The cooperative should wash the tubers in solution B and should keep containers out of the solar driers so that the temperature stays below 45 °C to stop losses of the coloured cell contents.

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	Both independent variables and the colour intensity are correct in the conclusion, and the recommendation is to use B and to stay at or below 45 °C.	3
Ae	One independent variable and the dependent variable are correct in both the conclusion and recommendation.	2
Al	One independent variable is correct in either the conclusion or the recommendation, with the dependent variable.	1
0	Conclusion or recommendation contradicts the candidate's table, or is absent.	0

END OF SCORING GUIDE